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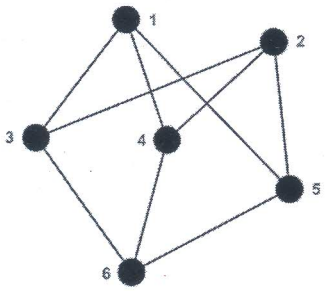
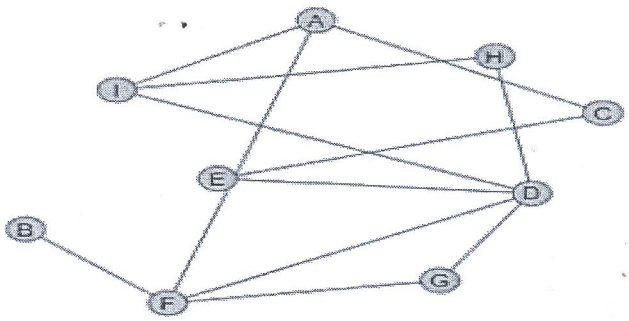
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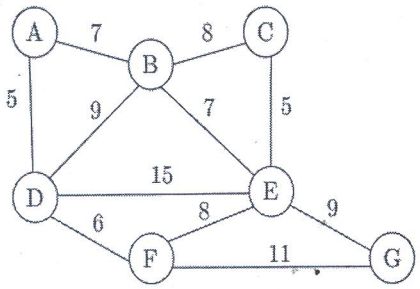
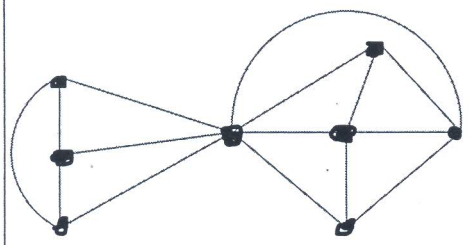
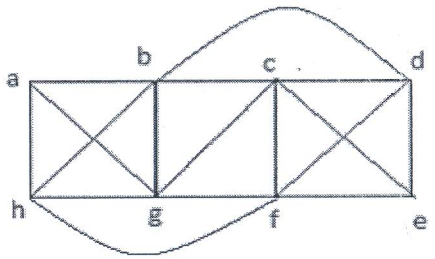
Programme: B. Tech (AI)/B. Tech (AI & DS)/B. Tech (AI & ML)/B. Tech (COMP)/B. Tech (CSBS)/B. Tech (CSE - CS)/B. TECH (CSE - DS)/B. Tech (EXTC)/B. Tech Intg. (CE)/B. Tech Intg. (IT)/B. Tech (IT)/MBA Tech (AI)/MBA Tech (Comp)/MBA Tech (IT)
Year: II / III / IV/Semester III/Semester V/Semester VII(Exam Year: 2025-2026)
Subject: Discrete Mathematics
Date: 24 Nov 2025 **Time: 10:00 am to 01:00 pm (03:00 Hrs.)**
Max. Marks: 100

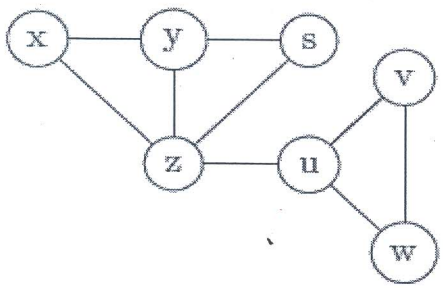
FINAL EXAMINATION (Acad. Year:**2025-2026**)

Instructions:
1. This question paper contains 4 pages.
2. Question No. 1 is compulsory.
3. Out of remaining questions, attempt any 4 questions.
4. In all 5 questions to be attempted.
5. All questions carry equal marks.
6. Figure in right hand side indicates full marks.
7. Answer to each new question to be started on a fresh page.
8. Assume Suitable data if necessary.

	Q.1		
CO-1; BL-1,2	a)	Let $\mathbb{Z}$ be a set of integers and let R be a relation on $A = \mathbb{Z} \times \mathbb{Z}$ defined by $(a,b)R(c,d)$ *if* $a + d = b + c$, then prove that R is reflexive and transitive.	4
CO-1 ; BL-1,2	b)	Using the truth table prove that following statement is a Tautology. $[(p \vee q) \wedge (p \longrightarrow r) \wedge (q \longrightarrow r)] \longrightarrow r$	4
CO-3; BL-3	c)	In a computer network of 30 computers, each computer is directly connected to at least 10 others. Show that at least two computers have the same number of direct connections.	4
CO-4; BL-3	d)	(i) Give an example of a regular, connected graph on six vertices that is not complete. (ii) Identify whether degree sequence (3, 2, 2, 1, 0) is graphic or not. If not then explain your answer, if yes draw graph.	2 2
CO-5; BL-2,3	e)	Let $G = Z_6^* = \{1,2,3,4,5\}$ under multiplication mod 6. Identify all elements that have an inverse. Find the inverse of 5.	4
	Q.2		
CO-3; BL-3	a)	Solve the recurrence relation $a_n = a_{n-1} + a_{n-2}$ with $a_1 = a_2 = 1$	6
CO- 4; BL-3,4	b)	(i) A simple graph has 12 vertices, and each vertex has a degree of either 3 or 5. If the number of vertices with degree 3 is twice the number of vertices with degree 5, determine the number of edges in graph G	6

		(ii) Check whether the following graph is a bipartite graph. Also write the corresponding adjacency matrix. 	
CO-1; BL-2,3	c)	If R is the relation defined on $\mathbb{Z}$ (set of integer), such that, $mRn \Leftrightarrow 4$ divides $(m-n)$. Prove that R is an equivalence relation.	8
	Q.3		
CO-2 ; BL-3	a)	Prove by mathematical induction that $n(n+1)(n+2)$ is a multiple of 6 for $n \geq 1$.	6
CO- 4; BL-5	b)	Determine the chromatic number of the graph given below using Welch Powell's algorithm and write steps of the algorithm. 	6
CO- 1; BL-2	c)	In a survey of 60 people, it was found that; 25 read Business India, 26 reads India Today, 26 reads Times of India, 11 reads both Business India and India Today, 09 reads both Business India and Times of India, 08 reads both India Today and Times of India, 08 reads none of the three. Then find (i) How many read all three? (ii) How many read exactly one?	8
	Q.4		
CO- 5; BL-4	a)	Prove that the set of non-zero real numbers $\mathbb{R}^*$ forms an abelian group with respect to the operation "$\circ$", where "$\circ$" is defined as $a \circ b = \frac{ab}{2}$, $\forall\, a, b \in \mathbb{R}^*$.	6
CO-2; BL-3	b)	Formalize the following and, by writing truth tables for premises and conclusion, determine whether the following argument is valid. *"If the global warming continues, then the temperature of the world's oceans will rise. If ocean temperatures rise, then hurricanes will become more intense. Therefore, if global warming continues, then hurricane will become more intense."*	6

CO- 3; BL-3	c)	Solve the recurrence relation: $a_n - 2a_{n-1} = 3(2^n)$.	8
	Q.5		
CO- 5; BL-4	a)	Let $\mathbb{R}$ be the set of real numbers and $\mathbb{R}^+$ be the set of positive real numbers. If $f: (\mathbb{R}, +) \to (\mathbb{R}^+, \cdot)$ is defined by $f(x) = e^x, \forall\, x \in \mathbb{R}$. Then show that f is a group isomorphism.	6
CO-1; BL-3	b)	Prove the following statement by contrapositive: For any integers m and n, if the product of m and n is even, then m is even or n is even.	6
CO- 4; BL-5	c)	Construct minimum spanning tree using Kruskal's algorithm. Also find minimum total weight. 	8
	Q.6		
CO-1; BL-2	a)	Define conditions for a function to be invertible. Explain why $f(x) = -3x^2 + 7$ is not invertible function from $\mathbb{R} \to \mathbb{R}$.	6
CO- 4; BL-2	b)	Find the edge connectivity and the vertex connectivity of the graphs:  	6
CO- 5; BL-3	c)	For $G = \{0,2,4,6\}$, i) Prepare composition table with respect to addition modulo 8 $(+_8)$. ii) Prove that it is an abelian group with respect to addition modulo 8 $(+_8)$. Is it cyclic? Justify your answer.	8
	Q.7		
CO- 2; BL-3	a)	Using laws of logic prove that $\sim(p \leftrightarrow q) \equiv (p \wedge \sim q) \vee (\sim p \wedge q)$	6
CO-5; BL-4	b)	Show that $(Q, +, \times)$ is a ring with unity, where Q is the set of rational numbers.	6

CO- 4; BL-5	c)	(i) Determine whether the given graph has an Eulerian path or an Eulerian circuit. If an Eulerian path or circuit exists, write one such path or circuit explicitly.  (ii) Provide an example of a graph that is Eulerian but not Hamiltonian, and an example of a graph that is Hamiltonian but not Eulerian. If such a choice is not possible, explain why.	8